Review: *When the Pipeline Is Wrong: Auditing the Evaluation of Scientific Agents*

Reviewer notes

Rating: about **8/10** (standard reviewer), **7/10** (harsh reviewer). Based on the full main text (pp. 1–7) and the appendix through p. 12.

Strengths

- **Clear, novel framing.** Truth, evidential support, decision consistency, and grader credit are scored separately, with a frozen oracle that recomputes confirmatory evidence. The "true but unsupported" and "supported but false" cells make the point concrete.
- **Careful statistics.** Pre-declared primary estimand, Holm-adjusted secondary contrasts, task-cluster bootstrap, leave-one-task-out refits, and Monte Carlo calibration of the oracle test (5.0% under the null).
- **Honest self-correction.** The provenance effect is stated to be positive by construction; v2 fault effects are called descriptive; v1 and v2 are not compared. The analysis showing 114 of 116 false discoveries come from two datasets is excellent.
- **Concrete, checkable findings**, e.g. 10 of 11 rejected traces were correct readings that broke the output format, and "one confirm call is not one test".
- **Figures 1 and 2 carry real information**; the appendix (formal DGP, calibration tables, instrumentation) is thorough.

Weaknesses

1. **The abstract is very dense.** Roughly eight numbers in nested sentences ("Of 19 traces credited only by an answer-only grader (+4.5 pp), 8 contradict … The other 11 … 10 of them …").
2. **The 42% headline overstates the problem.** The body splits it into 31.4% (several candidates in one model) and 10.9% (separate models), and concedes that a joint regression can be a single pre-specified test. Only the 10.9% is clearly a multiple-testing screen.
3. **"Enough to move the lowest-ranked model to first" is fragile**: it rests on 5 runs of one model and holds in 64% of bootstrap replicates. It should not be in the abstract.
4. **v2 fault effects are null.** All intervals include zero (Holm $p \geq 0.56$) with 4 models, one repeat, 40 tasks. The fault-injection contribution is therefore weak, and the significant v1 effects use coarser faults.
5. **Construct overload**: TRUTH, EVID, CONS, PROTO, replay-graded success, joint valid success, LEAK, SILENT, STALE, SEP_READ, INT_MASK, FRESH_CACHE, PROV, PROV_GRADE.

6. **v2 may be too constrained to measure much**: agents agree with the deterministic reference agent on 99.4% of decisions, and the paper concedes v2 certifies "hypothesis selection and decision discipline, not statistical programming".
7. **Narrow scope**: one synthetic linear data-generating process.
8. **Reference hygiene**: several entries use "et al." instead of full author lists (Zhu et al. 2025, Chen et al. 2025, Chan et al. 2025). FIRE-Bench (Wang et al. 2026, arXiv:2602.02905) should be verified against arXiv.

Recommended fixes, in priority order

1. Rewrite the abstract around about four numbers.
2. Lead with the 10.9% separate-model screening rate, not 42%.
3. Remove the ranking-flip claim from the abstract (keep it, with its caveats, in §5.2).
4. Add a short glossary table for the named constructs and fault operators.
5. Frame the v2 fault results explicitly as a power-limited null rather than a finding.
6. Complete all author lists and verify every reference against a primary record.